

Traditional Training

FP16/BF16 Computation

Fixed Precision for All Layers

Challenges

High Energy Consumption
High Security Risk

Continuous-bit ML Framework

WP1: Hybrid Architecture

WP2: Stochastic Sampling

WP3: Anti-Entropy Protocol

WP4: Validation

PASSION System

Hybrid Precision Architecture
(Stochastic + Deterministic)

Adaptive Bitstream Compression
(4-8 bit $\rightarrow$ 16-bit effective)

Outcomes

50-60% Energy Reduction
Enhanced Security Robustness